

Lista 01 - Resolucao

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Exercicio 1

a)

$$AB = B - A = (3 - 1, -1 - 2, 4 - (-1)) = (2, -3, 5)$$

$$BC = C - B = (-2 - 3, 0 - (-1), 2 - 4) = (-5, 1, -2)$$

$$CA = A - C = (1 - (-2), 2 - 0, -1 - 2) = (3, 2, -3)$$

b)

$$AB + BC + CA = (2, -3, 5) + (-5, 1, -2) + (3, 2, -3)$$

$$AB + BC + CA = (0, 0, 0)$$

c)

Resposta: percurso fechado A -> B -> C -> A; deslocamento resultante nulo.

Exercicio 2

a)

$$PQ = 2v = 2(-1, 3, 2) = (-2, 6, 4)$$

$$Q = P + PQ = (2, -1, 3) + (-2, 6, 4) = (0, 5, 7)$$

b)

$$QP = P - Q = (2 - 0, -1 - 5, 3 - 7) = (2, -6, -4)$$

$$|QP| = \text{raiz}(2^2 + (-6)^2 + (-4)^2) = \text{raiz}(56) = 2 \text{ raiz}(14)$$

Exercicio 3

a)

$$u \cdot v = (2, m, -3) \cdot (1, 4, 2)$$

$$u \cdot v = 2 \cdot 1 + m \cdot 4 + (-3) \cdot 2 = 4m - 4$$

b)

$$4m - 4 = 0$$

$$4m = 4$$

$$m = 1$$

Exercicio 4

a)

$$|a| = \text{raiz}(2^2 + (-6)^2 + 8^2) = \text{raiz}(104) = 2 \text{ raiz}(26)$$

$$|b| = \text{raiz}(7^2 + 1^2 + 4^2) = \text{raiz}(66)$$

b)

$$a \cdot b = 2 \cdot 7 + (-6) \cdot 1 + 8 \cdot 4 = 14 - 6 + 32 = 40$$

c)

$$\cos(\theta) = (a \cdot b) / (|a| |b|)$$

$$\cos(\theta) = 40 / (\text{raiz}(104) \text{ raiz}(66)) = 10 / \text{raiz}(429)$$

$$\theta = \arccos(10 / \text{raiz}(429)) \approx 61,13 \text{ graus}$$

$$\theta \approx 1,067 \text{ rad}$$

Exercicio 5

a)

$$u \times v = \begin{vmatrix} i & j & k \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 2 & 0 & 1 \end{vmatrix}$$

$$u \times v = (2 \cdot 1 - 0 \cdot 0, -(1 \cdot 1 - 0 \cdot 2), 1 \cdot 0 - 2 \cdot 2)$$

$$u \times v = (2, -1, -4)$$

b)

$$A_{\text{paralelogramo}} = |u \times v| = \text{raiz}(2^2 + (-1)^2 + (-4)^2) = \text{raiz}(21)$$

c)

$$A_{\text{triangulo}} = \text{raiz}(21) / 2$$

Exercicio 6

a)

$$a \times b = (1 \cdot 1 - 0 \cdot 1, 0 \cdot 0 - 1 \cdot 1, 1 \cdot 1 - 1 \cdot 0)$$

$$a \times b = (1, -1, 1)$$

$$|a \times b| = \text{raiz}(1^2 + (-1)^2 + 1^2) = \text{raiz}(3)$$

b)

$$|a| = \text{raiz}(2), |b| = \text{raiz}(2)$$

$$\text{sen}(\theta) = |a \times b| / (|a| |b|)$$

$$\text{sen}(\theta) = \text{raiz}(3) / (\text{raiz}(2) \text{raiz}(2)) = \text{raiz}(3) / 2$$

c)

$$h = \text{area} / \text{base} = \text{raiz}(3) / \text{raiz}(2) = \text{raiz}(6) / 2$$

Exercicio 7

a)

$$u \times v = (1 \cdot 1 - 0 \cdot 2, 0 \cdot 0 - 1 \cdot 1, 1 \cdot 2 - 1 \cdot 0)$$

$$u \times v = (1, -1, 2)$$

$$[u, v, w] = (u \times v) \cdot w = (1, -1, 2) \cdot (1, 0, 3)$$

$$[u, v, w] = 1 + 0 + 6 = 7$$

b)

$$V_{\text{paralelepipedo}} = |7| = 7$$

c)

$$V_{\text{prisma triangular}} = 7 / 2$$

$$V_{\text{tetraedro}} = 7 / 6$$

d)

$$v \times w = (2 \cdot 3 - 1 \cdot 0, 1 \cdot 1 - 0 \cdot 3, 0 \cdot 0 - 2 \cdot 1)$$

$$v \times w = (6, 1, -2)$$

$$|v \times w| = \text{raiz}(6^2 + 1^2 + (-2)^2) = \text{raiz}(41)$$

$$h = V / |v \times w| = 7 / \text{raiz}(41)$$

Exercicio 8

a)

$$n = u \times v = (0 \cdot 0 - 1 \cdot 1, 1 \cdot 1 - 1 \cdot 0, 1 \cdot 1 - 0 \cdot 1)$$

$$n = (-1, 1, 1)$$

b)

$$d \cdot n = 1(-1) + 2(1) + 1(1) = 2$$

$$|d| = \text{raiz}(6), |n| = \text{raiz}(3)$$

$$\cos(\alpha) = 2 / (\text{raiz}(6) \text{raiz}(3)) = \text{raiz}(2) / 3$$

$$\alpha \approx 61,87 \text{ graus}$$

c)

$$e \cdot n = 2(-1) + 1(1) + 1(1) = 0$$

$$\alpha = 90 \text{ graus}$$